

NAINA OS -- VOLUME 5

DESKTOP, MOBILE & COMPUTER INTERACTION FRAMEWORK (DMCIF v1.0)

"If Memory is the Mind, then Devices are the Body."

DOCUMENT METADATA

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Author Group:	Chief Systems Architect & Device Group
Flagship Feature:	Workspace Awareness Engine (WAE)
Core Focus:	Windows Win32 * Android ADB * Browser * OpenClaw * Hardware Drivers

This document contains formal engineering specifications for the NAINA OS Windows Runtime, Android Companion, Browser Automation, Application Plugins, Hardware Drivers, OpenClaw Loop, and Workspace Awareness Engine.

NAINA OS -- Desktop, Mobile & Computer Interaction Framework (DMCIF v1.0.0)

Volume 5: Host OS Runtime, Android Companion, Browser Automation, Hardware Drivers & Workspace Awareness

"If Memory is the Mind, then Devices are the Body."

Document Revision History

Date	Revision	Author	Description of Changes
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2026-07-31	v1.0.0	Chief Systems Architect	Release of Volume 5 -- Desktop, Mob
2026-07-29	v0.9.0	Systems Hardware Group	Specifications for Windows Runtime,

FLAGSHIP SUBSYSTEM: Workspace Awareness Engine (WAE)

1. Architectural Concept & Purpose

The Workspace Awareness Engine (WAE) elevates NAINA OS from a passive command runner into a context-aware operating environment. WAE continuously monitors screen layouts, active window handles, project working directories, process states, and peripheral connections.

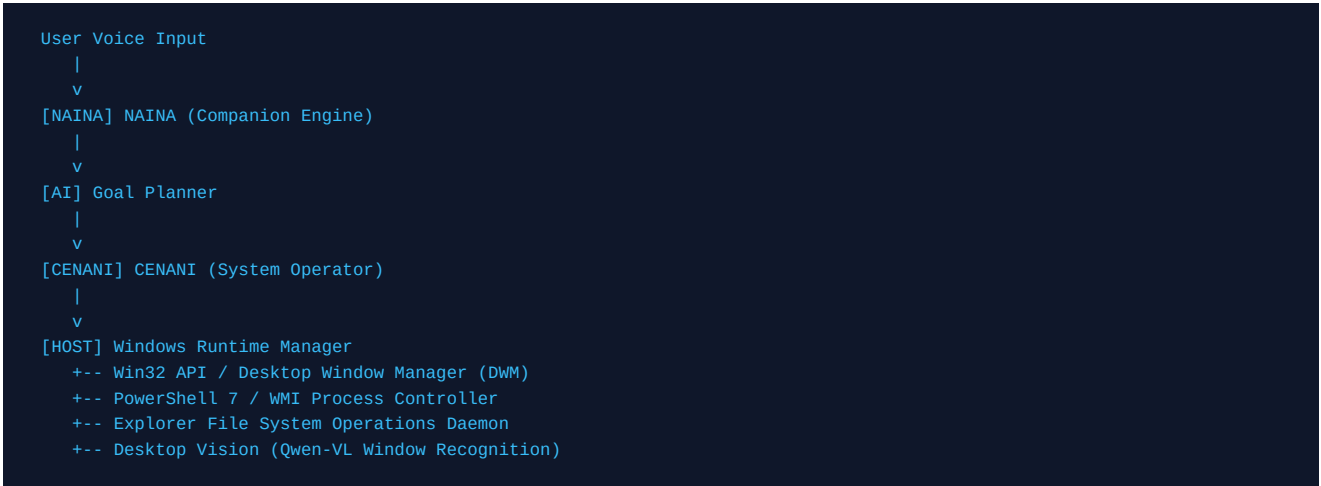
```
+-----+
|           WORKSPACE AWARENESS ENGINE (WAE)           |
+-----+
| MONITOR 1 (Primary Workstation) | MONITOR 2 (Monitoring Dashboard) | | | |
| +-----+ | +-----+ |
| | VS Code (NAINA OS Kernel Repo) | | Docker Desktop Build Logs | |
| +-----+ | +-----+ |
| +-----+ | +-----+ |
| | Chrome (GitHub PR Review) | | Terminal (Pytest Running) | |
| +-----+ | +-----+ |
+-----+
| WAE CONTEXT REASONING ENGINE |
| * Active Focus: Coding / Systems Engineering |
| * Active Project: NAINA OS Kernel (v1.0 Branch) |
| * Connected Hardware: Sony A58 Camera, Android Phone, USB Audio DAC |
| * Proactive Context Alert: "Docker build idle for 20m. Check logs?" |
+-----+
```

DOCUMENT 05.1 -- Windows Runtime Framework

1. System Architecture

The Windows Runtime Framework interfaces directly with Win32 APIs, PowerShell 7+, Windows Management Instrumentation

(WMI), and desktop process handles under CENANI's command loop.



2. Core Windows Subsystems & Capabilities

Subsystem Component	Direct Capability Description	System Interface
---	---	---
Application Manager	Discovers, launches, monitors, and	Process.GetProcesses(), Win32 API
Window Manager	Enumerates HWND handles, sets focus	EnumWindows(), SetForegroundWindow()
File System Daemon	Read, Write, Move, Compress (ZIP/7z)	Native Win32 File IO, vssadmin
Registry & Services	Inspects/modifies registry keys and	RegOpenKeyEx(), sc.exe
Terminal & Shell	Executes sandboxed PowerShell, CMD,	powershell.exe -NoProfile -Command
Performance Monitor	Real-time CPU, GPU (NVIDIA NVML), R	WMI / NVML C++ DLL Bindings
Application Plugins	OBS Studio (WebSockets), VS Code, D	Application Plugin Manifest API

3. Desktop Vision Engine

Instead of requiring exact executable names (e.g., "Launch chrome.exe"), Desktop Vision accepts spatial visual intents such as **"Switch to the browser window currently showing GitHub PR #42"**. It samples active HWND viewports, passes screenshots through Qwen-VL, and identifies target windows by visual content rather than process titles alone.

DOCUMENT 05.2 -- Android Companion Framework (NAINA Mobile)

1. System Architecture & Secure Channel

NAINA OS pairs seamlessly with Android mobile devices via ADB (Android Debug Bridge) or an encrypted local WebSocket bridge:

```
sequenceDiagram
    autonumber
    participant Desktop as [OS] NAINA Desktop Core
    participant Channel as [SEC] Encrypted TLS WebSocket / ADB
    participant Mobile as [PHONE] Android Runtime Agent
    participant PhoneOS as [ANDROID] Android Phone OS

    Desktop->>Channel: Send Encrypted Command (Intent: FETCH_SMS_AND_NOTIFS)
    Channel->>Mobile: Receive Payload
    Mobile->>PhoneOS: Query NotificationListenerService / Telephony API
    PhoneOS-->>Mobile: Return Device Telemetry (SMS, Battery, GPS)
    Mobile-->>Channel: Return Encrypted Response
    Channel-->>Desktop: Ingest into Session Memory
```

2. Mobile Capabilities Matrix

- * Communication: Read/Send SMS, manage Phone Calls, WhatsApp notification parsing (subject to OS permissions).
- * Sensors & Hardware: GPS Location queries, Battery charge telemetry, Camera frame capture, Mic audio streaming.
- * Connectivity: Wi-Fi status, Hotspot control, Bluetooth peripheral pairing, File transfer sync.

DOCUMENT 05.3 -- Browser Intelligence Framework

The Browser Intelligence Framework manages web interaction across Chrome, Firefox, Edge, Brave, and Arc:

```
User Prompt: "Research vector database indexing benchmarks and summarize in Obsidian."
|
v
[AI] AI Router & Goal Planner
|
v
[WEB] Browser Agent Subsystem
|
v
[HOST] OpenClaw Computer Use Engine
|
v
[PLAYWRIGHT] Playwright Headless Automation (Chrome / Firefox / Edge Drivers)
+-- Tab Management & Multi-Profile Support
+-- Cookie & Auth Session Persistence
+-- DOM Scraper & Markdown Text Extractor
```

DOCUMENT 05.4 -- Application Runtime Framework

Instead of building hardcoded, isolated agent scripts for every application, NAINA OS implements the Application Plugin Architecture:

```
{
  "application_name": "OBS Studio",
  "plugin_version": "1.0.0",
  "executable_match": "obs64.exe",
  "required_capabilities": ["CAP_NET_HTTP", "CAP_MEDIA_STREAM"],
  "supported_commands": [
    { "command_id": "obs_start_streaming", "mcp_tool": "obs_ws_start_stream" },
    { "command_id": "obs_switch_scene", "mcp_tool": "obs_ws_set_scene" }
  ]
}
```

DOCUMENT 05.5 -- Hardware Runtime & Device Drivers

The Hardware Runtime provides unified abstraction across physical computing peripherals:

- * Input Devices: Raw USB/Bluetooth Keyboard, Mouse pointer events, Touch digitizer.
- * Media Inputs: USB Webcams, Capture Cards, Microphone array beamforming.
- * Compute & Storage: CPU core frequency tuning, NVIDIA GPU CUDA streams, NVMe SSD SMART health.
- * Peripherals: Bluetooth audio DAC, USB printers, Multi-monitor Display Data Channel (DDC/CI).

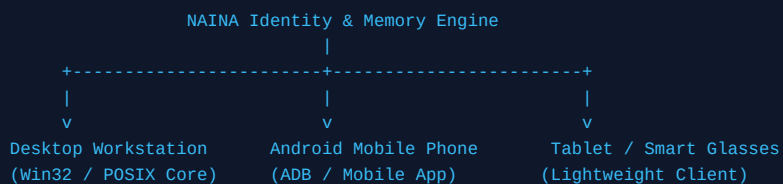
DOCUMENT 05.6 -- Computer Use Engine Execution Loop

The Computer Use Engine executes an autonomous vision-guided loop powered by OpenClaw:

```
flowchart TD
    S1[Step 1: Capture Viewport Screenshot] --> S2[Step 2: Qwen-VL Spatial UI Recognition]
    S2 --> S3[Step 3: Goal Planner Determines Action]
    S3 --> S4[Step 4: Safety Check ask_permission]
    S4 --> S5[Step 5: Execute Mouse Pointer Movement / Click]
    S5 --> S6[Step 6: Execute Keyboard Keystroke Buffer]
    S6 --> S7[Step 7: Capture Frame & Verify State Change]
    S7 --> S8[State Correct | Update Memory & Return Success]
    S7 --> S1[Shift Detected | UI Change Recovery Loop 3 Retries]
```

DOCUMENT 05.7 -- Multi-Device Synchronization

NAINA OS maintains a single unified identity across all hardware platforms:



- * One Identity: Single capability-gated security token across all devices.
- * One Memory: Real-time vector and Obsidian vault synchronization across nodes.

- * One AI Engine: Transparent model cascading distributing inference between local GPU rigs and mobile client nodes.

End of NAINA OS Desktop, Mobile & Computer Interaction Framework -- Volume 5 (Documents 05.1 - 05.7 & WAE)